

Tree Ring Climate Detective

Read the rings to reconstruct the climate events a tree lived through.

Win condition: Most accurate climate-event inferences with claim-evidence-reasoning.

THE SCIENCE YOU ARE USING

Rings as proxy data. A tree adds one ring per year. Wide rings mean a good growing season; narrow rings mean stress like drought, cold, or crowding. Because the tree cannot record numbers, the rings are a proxy: an indirect record of past climate.

Claim, evidence, reasoning. You infer events from ring patterns, then justify each call with the specific rings that support it. Scars, sudden narrowing, and runs of wide rings are your clues. Strong inferences cite the evidence.

HOW TO BUILD AND RUN IT

- 01 Learn the ring code.** Wide equals a favorable year, narrow a stress year (drought, cold, or crowding), scar a fire the tree survived. A ring is a proxy, not a thermometer.
- 02 Read the sequence.** Work along the rings from center to bark, noting each pattern and the year it maps to.
- 03 Infer the events.** Match patterns to likely events: drought, fire, a crowded stand, a recovery.
- 04 Cite your evidence.** For each inference, point to the exact rings that justify it on a claim-evidence card.
- 05 Synthesize the story.** Combine your inferences into one short climate history for the tree.

Safety: Sand wood cookies smooth if used. No cutting tools with students. Allergy: Wood dust sensitivity note if using real samples.

MATERIALS

Printed ring cards	one set per team (4 teams, plus 1 spare)
Answer boards	one per team (4, plus 1 spare)
Wood cookies (optional)	one per team if used (sanded), printed cards are the primary medium
Timers	one per team (4) or the projected room timer
Claim-evidence cards	a stack per team (about 5 to 7 each)

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (which ring pattern is the clearest sign of a hard year, and why): _____

Our climate history for the tree, oldest ring to most recent: _____

The events we inferred (drought, fire/scar, crowding, recovery): _____

The exact rings (which years and patterns) that are our evidence for each event: _____

DEFEND YOUR READING

Which ring pattern was easiest to misread, and how did you resolve it? _____

If you read it again, what is the ONE inference you would re-check, and with which rings? _____

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Correct inferences	40
Evidence justification	20
Speed	20
Final synthesis	20
Total	100